A-A-50434
15 AUGUST 1990

COMMERCIAL ITEM DESCRIPTION

PATCH REPAIR EMERGENCY WATER ACTIVATED DAMAGE CONTROL

The General Services Administration has authorized the use of this commercial item description.

Abstract. This description covers an Emergency Water Activated Repair Patch (EWARP) which is used by the U.S. Navy for interim leak stopping on metallic pipes. The EWARP consists of a polyurethane resin impregnated fiberglass cloth material.

Salient characteristics:

DESCRIPTION: The EWARP shall be produced in two sizes:

Size 1 - 2 (± 1/4) wide by 5 feet long.
Size 2 - 4 (± 1/4) wide by 15 feet long.

The EWARP will be used in repairing fresh water, salt water, compressed air, hydraulic fluid, and lubricating oil piping systems up to 2-1/2 inches in diameter with pressures not to exceed 150 pounds per square inch gauge (lb/in^2), and temperatures not to exceed 300 degrees Fahrenheit (^{o} F). The EWARP shall be a one-part, ready to use material, which requires no pre-mixing or application of additional chemical compounds. The EWARP yarn shall be fibrous glass, and have a thickness of 0.020 (± 0.007) inches. The size 1 and size 2 impregnated cloth shall one continuous length and not spliced. The size 1 and size 2 EWARP shall be wound on a minimum 3/8-inch diameter core and packaged as specified. The EWARP shall be thoroughly impregnated with sufficient water activated resin to fulfill its purpose. The EWARP shall meet the performance requirements as defined in this Commercial Item Description.

Beneficial comments, recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 55Z3, Department of the Navy, Washington, D.C. 20362-5101 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this Document or by letter.

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1. Accessories:
(a) Latex Examination Gloves: Each pouch containing an EWARP, shall also contain a pair of latex gloves. The gloves shall be made of latex, 0.005 thick (+ 0.5 mil), and 9 inches (± 0.5 inches) from the cuff to the middle of the third finger in length. Size supplied shall be 10 - 11.

(b) Visitor's Specs: Each case of EWARP shall contain one (1) pair of Visitor's Specs. The Visitor's Specs shall be of clear colorless plastic, and typical commercial quality. They shall have integral side shields and they shall satisfactorily fit and function when worn over prescription eye glasses typically in use. The Visitor's Specs shall be individually packaged in clear plastic packaging material.

2. Shelf Life: The EWARP shall have a shelf life as packaged for a minimum period of 24 months.

3. Performance Requirements: Prior to the application of EWARP, the piping system shall be at an ambient temperature in the range of 60 to 90 degrees fahrenheit. No media shall be flowing through the piping section being repaired. EWARP shall be used independent of augmenting materials such as boiler sealing compounds or rubber gasket material. The pipe surface shall not be cleaned, other than to remove loose rust and scale and jagged or protruding edges around the rupture. The rupture size shall not exceed 0.75 square inches for compressed air, hydraulic fluid and lubricating oil, and 3 square inches for fresh and salt water. The EWARP shall be removed from the hermetically sealed pouch and presoaked in fresh or salt water for not more than 20 seconds. No more than one (1) pack of size 1 and two (2) packs of size 2 EWARP shall be applied to the pipe. The EWARP shall solidify in not less than 5 minutes and not more than 15 minutes within this temperature range. The EWARP shall be allowed to cure for 30 minutes. After 30 minutes, the EWARP shall be cured to the point that full pressure of 150 lb/in^2 can be applied to the pipe without leakage within this same temperature range.

Quality assurance. The EWARP shall be of the highest quality and shall be free from contaminants, foreign, or extraneous matter which will affect the performance of the EWARP. Unless otherwise specified in the contract, the contractor is responsible for the performance of all quality inspections of the EWARP. The government reserves the right to perform any First Article Inspections and require First Article Testing to assure that supplies conform to prescribed requirements.

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1. Health Requirement: When the EWARP is used in combination with air, fresh water, salt water, hydraulic fluid, or lubricating oil, its fumes shall have no adverse effect on the health of personnel when used for its intended purpose. Neither should the EWARP present any unusual problems that would require more protective equipment than Visitor's Specs and gloves.

2. Safety Requirement: Material Safety Data Sheets (MSDS) in accordance with the Occupational Safety and Health Administration (OSHA) Hazard Communication Standard 29 CFR 1910.1200 shall be provided to point of delivery for each case of EWARP. Cases not containing an MSDS will be returned.

3. TDI and MDI Certification: Prior to award of the contract, the manufacturer shall provide certification that Toluene Diisocyanate (TDI) has not been used in the formulation of the EWARP, and that the airborne levels of diphenylmethane diisocyanate (MDI) released during normal use of the EWARP do not exceed 0.02 PPM (or current permissible exposure criteria) ceiling value as stipulated by Title 29 Code of Federal Regulations, Part 1910.1000. Certification that TDI is not present, and the MDI airborne levels shall be made by a board certified Industrial Hygienist, based upon laboratory results obtained from an analytical laboratory accredited by the American Industrial Hygiene Association. The method of sampling and analysis shall be approved by the National Institute for Occupational Safety, and Health. A standard amount of one (1) pouch of size 1 and two (2) pouches of size 2 EWARPs must be applied for each sample, and a sufficient number of samples shall be taken to document on a statistical basis that the safe exposure limits of MDI do not exceed 0.02 PPM (or current permissible exposure criteria) as a ceiling value in the breathing zone. If the level of MDI given off by the EWARP exceeds the 0.02 PPM (or current permissible exposure criteria) ceiling value in the breathing zone, the EWARP shall be rejected.

4. First Article Tests. The Government reserves the right to require First Article Testing of EWARP prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract. The tests shall be conducted as follows:

(a) EWARP application procedure: The EWARP shall be applied to the pipes in accordance with the Manufacturer's instructions, and in the manner and under the conditions specified in the performance requirements.

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(b) Fresh Water, and Salt Water Tests Procedure: A test shall be conducted for each media. The test pipes shall be a 2.5 inch schedule 40 pipe, 18 to 24 inches long, fitted with a removable cap at each end. There shall be one 1/2 inch wide by 6 inch long hole drilled in each pipe specimen. Thirty (30) minutes after application of EWARP, each pipe specimen shall be filled with the required media and pressurized to a static pressure of 150 lb/in^2, and maintained at 150-140 lb/in^2 for 30 minutes.

Pass / Fail Criteria: The Fresh Water and Salt Water test specimens shall show no signs of leakage, or pressure droppage below 140 PSIG for a period of 30 minutes.

(c) Compressed Air, Hydraulic Fluid (MIL-H-83282), and Lubricating Oil, Tests Procedure: A test shall be conducted for each media. The test pipes shall be a 1 inch schedule 40 pipe, 18 to 24 inches long, fitted with a removable cap at each end. There shall be one 1/2 inch wide by 1-1/2 inch long hole drilled in each pipe specimen. Thirty (30) minutes after application of EWARP, each pipe specimen shall be filled with the required media and pressurized to a static pressure of 150 lb/in^2, and maintained at 150-140 lb/in^2 for 30 minutes. Next the test specimens containing the Hydraulic Fluid, and Lubricating Oil, shall be un-pressurized but remain filled, and left to stand for a period of not less than 5 hours, after such time, the test specimens shall be re-pressurized to a static pressure of 150 lb/in^2 and maintained at 150-140 lb/in^2 for 30 minutes.

Pass / Fail Criteria: The Hydraulic Fluid, and Lubricating Oil shall not chemically react with the EWARP material and cause leakage or pressure droppage below 140 lb/in^2 for a period of 30 minutes.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Contractor shall maintain records of First Article Tests. The Government reserves the right to require proof of quality conformance and First Article Testing prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with public law 94-580 to the maximum extent practical.

Preservation, packaging, packing, and marking.

1. Preservation: Each EWARP shall be snugly wound on a core minimum 3/8 inch in diameter, and then hermetically sealed in an aluminum foil bag or pouch.

2. Packaging: The preserved EWARP and a pair of latex gloves shall be unit packaged in a minimum 0.004 inch thick transparent virgin polyethylene bag and sealed. Staples shall not be used.

3. Packing: Ten (10) EWARPs of the same size (size 1 or size 2) shall be packed in separate cases (shipping containers) for delivery to the contracting activity. Each case shall include one (1) pair of Visitor's Specs, a copy of the MSDS, and a copy of the MDI and TDI Certifications.

4. Labeling and Marking: In addition to any special marking specified in the contract or order, each bag or pouch, unit package and exterior shipping container (case) shall be labeled or permanently marked as follows:

a. Bags and Pouches.

Item Description:
EMERGENCY WATER ACTIVATED REPAIR PATCH (EWARP)

Size:
2 IN. X 5 FT. or 4 IN. X 15FT. (as applicable)

National or NATO Stock Number:
When no NSN/NATO stock number is available, the FSCM appropriate to the part number shall be shown followed by the manufacturers's reference/part number (MFR/PN). The letters "FSCM" shall be used to identify this information.

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Instructions for use:
In addition to the manufacturers instructions, the words " When used aboard navy ships, see the Naval Ships Technical Manual (NSTM) 079 Volume 2 (NAVSEA S9086-CN-STM-020), Federal Stock Number 0901-LP-079-0020 for addition information."
Shelf Life Expiration date: (twenty four (24) months from date of manufacture)
Precautionary markings:
WARNING: Do not use this product on potable water inlet lines or fuel systems. Do not use on systems whose maximum operating pressures exceed 150 PSIG. Do not use on systems whose operating temperatures exceed 300 degrees ^{o} F. Wear enclosed protective gloves and Visitor's Specs at all times when handling or working with this material.
Avoid contact with unprotected areas of skin and eyes. In case of contact with skin, swab lightly with alcohol or other approved cleaning fluid.
In case of contact with the eyes, flush eyes with water for 15 minutes and get immediate medical attention.

b. Unit packages:
Unit packages shall contain permanent printed markings or a printed label containing the same information required for an EWARP bag or pouch.

c. Shipping containers:
NSN/NATO stock number.
FSCM and MFR/PN.
When no NSN/NATO stock number is available or when specified in the contract or order.
Item description.
Quantity and unit of issue.
Gross weight and cube.
Proper shipping name(s) and identification numbers.
NATO, United Nations (UN number) for hazardous items. The proper shipping name shall be distinct and separate from the item description and will be placed below the gross weight and cube area.
Hazardous material labels.
As required/applicable in accordance with the Department of Transportation hazardous material regulation, Title 49 parts 101 through 179.

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Notes.

1. The Government may elect not to evaluate a bid sample from an apparent low bidder for award for this solicitation if a determination of urgency is made by the contracting officer. The bid sample would then be evaluated prior to future solicitation for comparable items.

2. Occupational Safety and Health Administration (OSHA) Hazard Communication Standard 29 CFR 1910.1200 is available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.

MILITARY INTERESTS:

Military Coordinating Activity

Navy - YD

Custodians

Army - ME

Review Activity

Army - MD

User Activity

Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS

PREPARING ACTIVITY:

NAVY - SH

Project 4730-0011

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Brand name products believed to be acceptable, based upon information received from the company, and of current manufacture is as follows:

SYNTHO-GLASS™ (Industrial Grade) from Neptune Research Inc., 2611 Old Okeechobee Road, Suite #3, West Palm Beach, Florida 33409. Commercial telephone number is (407) 683-6992. Federal Cage No. 67377.

STOP-IT™ from InduMar Products, Inc, Post Office Box 770008 Houston, Texas 77215-0008 Commercial telephone number is (713) 977-4100

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE		

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact: Mr. Charles Grotenroth	b. TELEPHONE (Include Area Code) (1) Commercial 703-602-7217	(2) AUTOVON 8-322-7217
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command (5523) Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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